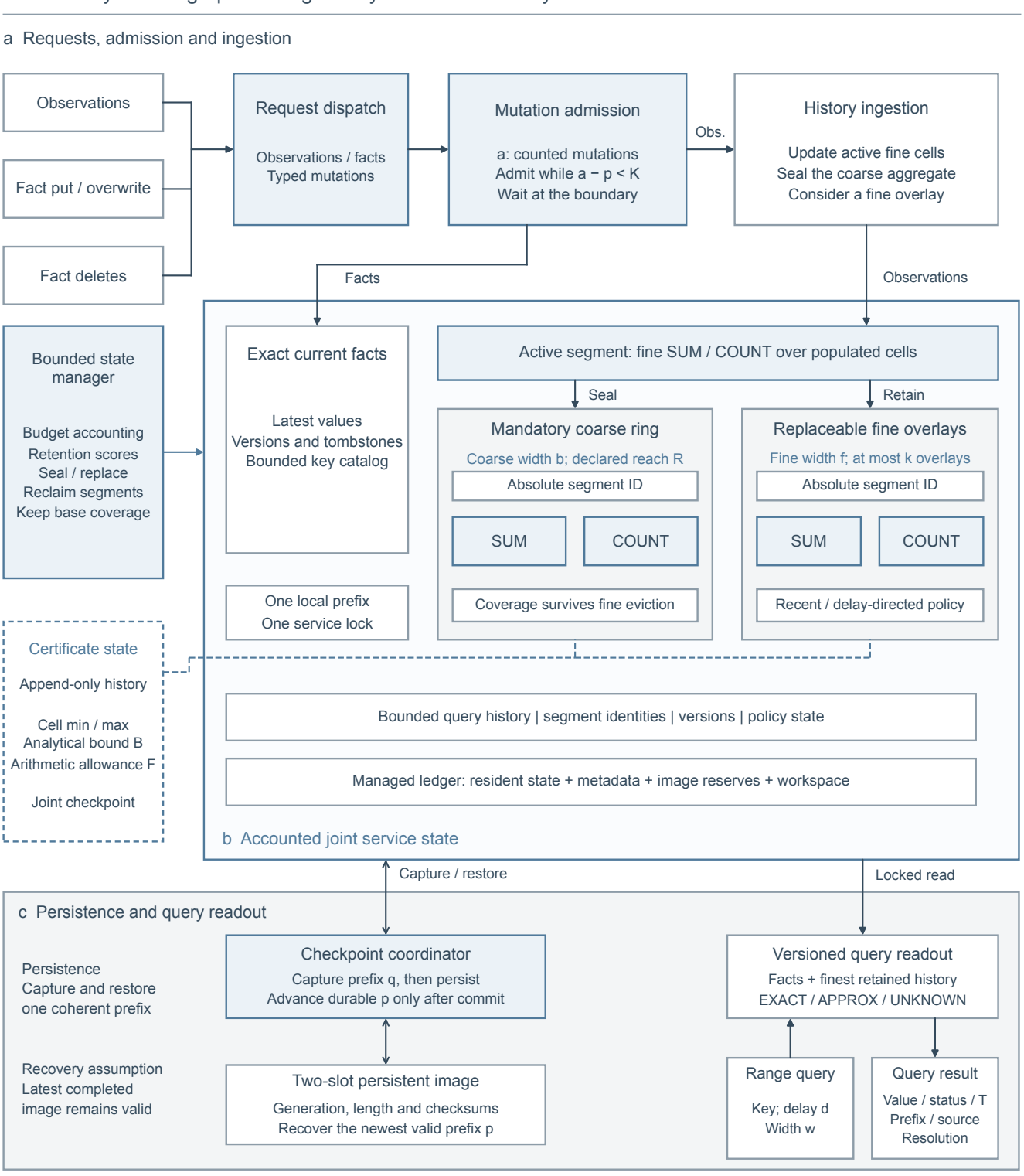
MT-Memory: coverage-preserving history and local recovery
a Requests, admission and ingestion
Observations
Fact put / overwrite
Fact deletes
Request dispatch
Observations / facts
Typed mutations
Mutation admission
a: counted mutations
Admit while a − p < K
Wait at the boundary
Obs.
History ingestion
Update active fine cells
Seal the coarse aggregate
Consider a fine overlay
Facts
Observations
Bounded state manager
Budget accounting
Retention scores
Seal / replace
Reclaim segments
Keep base coverage
Exact current facts
Latest values
Versions and tombstones
Bounded key catalog
Active segment: fine SUM / COUNT over populated cells
Seal
Retain
Mandatory coarse ring
Coarse width b; declared reach R
Absolute segment ID
SUM
COUNT
Coverage survives fine eviction
Replaceable fine overlays
Fine width f; at most k overlays
Absolute segment ID
SUM
COUNT
Recent / delay-directed policy
One local prefix
One service lock
Certificate state
Append-only history
Cell min / max
Analytical bound B
Arithmetic allowance F
Joint checkpoint
Bounded query history | segment identities | versions | policy state
Managed ledger: resident state + metadata + image reserves + workspace
b Accounted joint service state
Capture / restore
Locked read
c Persistence and query readout
Persistence
Capture and restore
one coherent prefix
Recovery assumption
Latest completed
image remains valid
Checkpoint coordinator
Capture prefix q, then persist
Advance durable p only after commit
Two-slot persistent image
Generation, length and checksums
Recover the newest valid prefix p
Versioned query readout
Facts + finest retained history
EXACT / APPROX / UNKNOWN
Range query
Key; delay d
Width w
Query result
Value / status / T
Prefix / source
Resolution
Solid: service execution. Dashed: certificate-to-cell associations in joint state.